Longitudinal Data Analysis
Assignment 3: Incomplete Data

Luis César Herrera Paltán, Daniel Sanson,
Maryam Shojaei Shahrokhabadi, Nele Verhulst

12 January 2026

1 Introduction

This report looks further into the longitudinal data of a study on Alzheimer's disease, focusing on missingness in the data. The study followed 1253 patients starting with baseline measurements at their first diagnosis and then annually over six years. Baseline patient data included in the analyses were age (AGE), sex (SEX), body mass index (BMI), activities of daily living (ADL), and a possible biomarker ABPET. Two cognitive scale variables were measured at each of the seven time points: the Brief Psychiatric Rating Scale (BPRS) and the Clinical Dementia Rating Sum of Boxes (CDRSB). The first report focused on how the continuous outcome BPRS changed over time, while the second report looked into a dichotomized version of the CDRSB over time.

This third report focuses on whether and how missingness in the dataset affected the previous analyses. Several mechanisms exist for missingness in data: in an MCAR (missing completely at random) missingness mechanism, the dropout is independent of both unobserved and observed data; if, conditional on the observed data, the dropout is independent of the unobserved measurements, the mechanism is MAR (missing at random); otherwise it is termed MNAR (missing not at random) [1]. This report first explores the missingness in the data. Then the continuous BPRS and the binary CDRSB outcome are analyzed assuming MAR. Additionally, for the continuous BPRS outcome, a sensitivity analysis explores the potential impact of deviations from MAR on the conclusions of the analysis.

2 Statistical methods

2.1 Data preparation and exploration

For the exploration of missingness, several baseline covariates were categorized. Age groups were defined as (1) ≤ 67, (2) $68 - 77$, and (3) ≥ 78. For BMI, common categories of normal ($18.5 \leq BMI \leq 24.9$), overweight ($25 \leq BMI \leq 29.9$) and obese ($BMI \geq 30$) were used. Multiple questionnaires exist that assess performance in ADLs of adults living with Alzheimer's disease and scale and scoring methods vary [2]. It was not clear from the dataset which questionnaire or scale was used, so four categories were determined ad hoc with ADL (1) between 0 and 4, (2) between 5 and 9, (3) between 10 and 14, (4) between 15 and 20. For Amyloid-Positron Emission Tomography (ABPET) at baseline, measured as the standardized uptake value ratio of a specific radioactive tracer, two groups were used with values (1) < 2.5, (2) ≥ 2.5.

The CDRSB data were dichotomized to be 0 if CDRSB ≤ 10 (so for less advanced dementia) and 1 if CDRSB > 10 (so more advanced dementia), referred to as low and high CDRSB, respectively, in this report. The dichotomized CDRSB baseline data was used in the exploration of missingness. The dichotomized CDRSB was also analyzed as longitudinal outcome variable in the second part of this report. In the longitudinal analyses, all covariates, except sex, were continuous.

Missingness patterns were first examined by summarizing the sequence of observed and missing measurements across the seven scheduled visits (Year 0 to Year 6) for each subject, with observed measurements coded as O and missing measurements as M. Then, for the baseline covariates, the number of patients per group and their proportion of the total was examined at each time point to gain further insight into dropout

patterns. For the BPRS and CDRSB longitudinal outcomes, the mean evolution over time was visualized per dropout pattern.

2.2 Random-effects model for BPRS score

In likelihood-based inference, MAR is ignorable, which means that inferences about the measurement mechanism can be made without explicitly addressing the missingness mechanism [3]. Hence, a linear mixed-effects model (LMM), valid under MAR, was chosen for the analysis of the continuous BPRS outcome over time for all 1253 patients. The first report's analysis was adjusted, to ensure consistency among covariates in the current report. Baseline covariates included sex, age, BMI, baseline ABPET and CDRSB, all continuous variables except sex. Random effects (intercept and slope) were included, since they reflect how much the subject-specific profiles (i.e. outlying individuals) or groups of individuals evolved differently in time [4]. The best model was determined using a step-by-step approach to identify the most parsimonious specification. Covariance structure was selected via likelihood ratio tests, using restricted maximum likelihood (REML), to compare Unstructured, ARH(1), and Toeplitz. The full model was formulated as:

$$\begin{aligned}\text{BPRS}_{ij} = \beta_0 + b_{0i} + \beta_1\text{Sex}_i + \beta_2\text{Age}_i + \beta_3\text{BMI}_i + \beta_4\text{CDRSB}_i + \beta_5\text{ABPET}_i + \\ + (b_{1i} + \beta_6\text{Sex}_i + \beta_7\text{Age}_i + \beta_8\text{BMI}_i + \beta_9\text{CDRSB}_i + \beta_{10}\text{ABPET}_i) \cdot \text{Year}_{ij} + \epsilon_i\end{aligned}$$

where:

- b_{0i}, b_{1i}: Subject-specific random intercept and slope, assumed to follow

$$\mathbf{b}_i = (b_{0i}, b_{1i})^\top \sim N(\mathbf{0}, \mathbf{D}).$$

- ϵ_i: Error component $\sim$ N(0,σ^2)
- βs: Population-average fixed effects for covariates and their linear interactions with time.
- Sex: Binary variable with Male as reference group (sex=0).
- $Age_i, BMI_i, CDRSB_i, ABPET_i$: Baseline continuous covariates for i-th patient.

2.2.1 Sensitivity analysis

Ignorable models are valid under the MAR assumption; however, MNAR, where missingness depends on unobserved data, may be present, and MAR cannot be distinguished from MNAR based on observed data alone. Therefore, a sensitivity analysis was conducted within the pattern-mixture model (PMM) framework, implemented through multiple imputation [1]. In this framework, the available-case missing value (ACMV) approach reflects the MAR assumption. In contrast, the complete-case missing value (CCMV) and neighboring-case missing value (NCMV) restrictions are alternative identifying restrictions within the pattern-mixture framework. They are not implied by the MAR-based ACMV restriction and are therefore used to assess the sensitivity of the analysis to departures from the benchmark assumption.

Multiple imputation was performed within the pattern-mixture model framework to assess the sensitivity of the results to departures from the MAR assumption. Preliminary analyses using 5 imputations showed results comparable to those with 10 imputations. To improve the stability of the pooled estimates and standard errors, the final sensitivity analysis was based on 10 imputations, with results combined across imputations using Rubin's rules [5]. For each dropout pattern, a model was fitted to the observed data, and conditional distributions of the unobserved measurements were specified according to the chosen identifying restriction: ACMV, CCMV, or NCMV. Imputed values were drawn to create completed datasets, each of which was analyzed using the intended statistical model.

2.3 Marginal model for CDRSB

Since Generalized Estimating Equations (GEEs) are based on a frequentist inference paradigm, they are valid under the more restrictive MCAR assumption. There are two main alternatives that are both valid under MAR: weighted GEEs (WGEEs), and multiple imputation followed by GEE (MI-GEE)[3]. MI-GEE

requires imputing missing outcomes for all subjects until the end of the study. A meta-analysis found that after diagnosis, the mean survival time for people with Alzheimer's disease was 5.8 years (SD 2.0) [6]. Thus, even though there was no information included with the dataset about the reason for missingness, it can reasonably be assumed that at least part of the missingness is due to patients dying during the 6-year time span of the study. Therefore, the interpretation of imputed values related to disease progression, like CDRSB, beyond death would be difficult and the focus was on the WGEE approach for interpretation, with the MI-GEE results only included for comparison.

The general idea of WGEE is to weight each subject's measurements in the GEEs by the inverse probability that a subject drops out at that particular measurement occasion [3]. The marginal mean model for the longitudinal binary outcome Y_{ij} ($Y_{ij} = 1$ if CDRSB > 10, and 0 otherwise) used in WGEE was formulated as:

$$\begin{aligned}\text{logit}\left[P(Y_{ij}=1|Year_{ij},Sex_i,Age_i,BMI_i,ADL_i,ABPET_i)\right] = \\ \beta_0+\beta_1\text{Sex}_i+\beta_2\text{Age}_i+\beta_3\text{BMI}_i+\beta_4\text{ADL}_i+\beta_5\text{ABPET}_i \\ +(\beta_6+\beta_7\text{Sex}_i+\beta_8\text{Age}_i+\beta_9\text{BMI}_i+\beta_{10}\text{ADL}_i+\beta_{11}\text{ABPET}_i)\cdot Year_{ij}\end{aligned}$$

where:

- $Year_{ij}$: is the time point at which the jth measurement is taken for the ith subject.
- Sex: Binary variable with Male as reference group (sex=0).
- $Age_i, BMI_i, ADL_i, ABPET_i$: Baseline continuous covariates for i-th patient.
- The βs: Population-averages fixed effects for covariates and their linear interactions with time.

An unstructured working correlation was adopted for the WGEE analysis, as it best captured correlation in the data. To account for informative dropout, inverse-probability weights were constructed based on the probability of dropout at a given time, conditional on the patient remaining in the study, their past measurements, and relevant covariates including sex, BMI, age, ADL, baseline ABPET, and the most recent outcome. For the most recent outcome, the original continuous CDRSB was used rather than the dichotomized version, to get the most precise weights.
The research question about the effects of ADL and ABPET on the longitudinal profiles of CDRSB at the population level was re-evaluated by testing whether the coefficients $\beta_4, \beta_5, \beta_{10}, \beta_{11}$, representing baseline and time effects of ADL and ABPET, differed significantly from zero. The results of WGEE were compared with a standard GEE analysis to evaluate the impact of accounting for dropout. Additionally, the WGEE results were compared to those of MI-GEE, with 10, 15 and 20 imputations to check the needed number of imputations. The same model and working correlation (unstructured) were used in the different GEE analyses.

2.4 Random-effects model for CDRSB

The longitudinal binary CDRSB outcome Y_{ij} was analyzed at the subject level with a Generalized Linear Mixed Model (GLMM). This model is valid under MAR assumption[3], making the missingness mechanism ignorable and eliminating the need for imputation. A logistic mixed-effects model with linear time trends was fitted, including sex, age, BMI and ABPET as baseline covariates, and accounting for within-subject correlation through random effects (slope and intercept). The model was specified as follows:

$$\begin{aligned}\text{logit}\left[P(Y_{ij}=1|Year_{ij},Sex_i,Age_i,BMI_i,ADL_i,ABPET_i,b_{0i},b_{1i})\right] = \\ \beta_0+b_{0i}+\beta_1\text{Sex}_i+\beta_2\text{Age}_i+\beta_3\text{BMI}_i+\beta_4\text{ADL}_i+\beta_5\text{ABPET}_i \\ +(b_{1i}+\beta_6+\beta_7\text{Sex}_i+\beta_8\text{Age}_i+\beta_9\text{BMI}_i+\beta_{10}\text{ADL}_i+\beta_{11}\text{ABPET}_i)\cdot Year_{ij}\end{aligned}$$

where:

- b_{0i}, b_{1i} are the subject-specific random intercept, and slope, assumed to follow

$$\mathbf{b}_i = (b_{0i}, b_{1i})^\top \sim N(\mathbf{0}, \mathbf{D}).$$

- The β coefficients represent subject-specific fixed effects for the corresponding covariates and their interactions with linear time.
- All variables are defined as in Section 2.3.

Model parameters were estimated by maximum likelihood using adaptive Gaussian quadrature (AGQ) to handle the random effects. In the second report, different numbers of quadrature points were compared and parameter estimates and standard errors were very similar for 10 and 20 quadrature points. Therefore, the results for 20 quadrature points were presented in this report. As in the WGEE analysis, the primary interest was to assess whether baseline and time-varying effects of ADL and ABPET affected the probability of having a CDRSB score greater than 10 over time.

3 Results and discussion

3.1 Data exploration

Data for a combination of patient and time point were either complete or completely missing, so missingness patterns were identical for all variables. There were only monotone missingness patterns (Table 1). Observations were complete through Year 6 for 40.8% of patients, while the remaining patients were divided approximately evenly over the other six dropout times (Table 1).

The missingness mechanism could not reasonably be assumed to be MCAR, as the probability of missingness seemed to be associated with several of the baseline covariates (Table 3). At baseline, the number of male and female patients was approximately the same, while in Year 6 only one third of the patients was female. Patients $\geq$78 years old at baseline made up almost a quarter of the study group at first, but their proportion declined steadily until less than 2% in Year 6. Patients with high ABPET at baseline made up about one third of the patient group, but had more drouput than the group with low ABPET at baseline, with only 28 of the high baseline ABPET patients left in the study at Year 6. Dropout rate was similar for the group with low and high CDRSB at baseline, with about 30% of patients having CDRSB > 10 at baseline and those patients continued to make up about 30% of the patient group in Year 6.

Looking at the mean BPRS and CDRSB profiles by dropout pattern, at each time point the group dropping out the next year had the most advanced dementia, as rated on both cognitive scales (Figure 1). In some cases, the difference in score with the next dropout group was small – like in Year 2 when patients that dropped out the next year had only slightly higher mean BPRS than patients that continued for one more year– while in other instances the difference was large, like in Year 5 (Figure 1).

Given the longitudinal design, the monotone dropout structure, and the dependence of dropout on observed measurements, the missing data mechanism was assumed to be MAR, whereby the probability of missingness can depend on observed data but not on unobserved outcomes. An MNAR missingness mechanism with dropout dependent on the unobserved outcome values could not be ruled out, so a sensitivity analysis will be done to evaluate how robust the conclusions are for the continuous outcome.

Table 1: Missingness pattern for the Alzheimer dataset (O = observed; M = missing)

Measurement occasion								
Y0	**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Count**	**%**
Completers								
O	O	O	O	O	O	O	511	40.78
Dropouts								
O	O	O	O	O	O	M	141	11.25
O	O	O	O	O	M	M	125	9.98
O	O	O	O	M	M	M	130	10.38
O	O	O	M	M	M	M	107	8.54
O	O	M	M	M	M	M	92	7.34
O	M	M	M	M	M	M	147	11.73

Table 2: Retention over time by subgroups of sex, age, BMI, ADL, and baseline ABPET and CDRSB. Percentages represent the proportion of the total sample retained at each time point.

Variable	**Group**	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**
Sex	Male	616 (49.2%)	583 (52.7%)	551 (54.3%)	506 (55.8%)	463 (59.6%)	409 (62.7%)	341 (66.7%)
	Female	637 (50.8%)	523 (47.3%)	463 (45.7%)	401 (44.2%)	314 (40.4%)	243 (37.3%)	170 (33.3%)
Age group	≤67 years	318 (25.4%)	318 (28.8%)	318 (31.4%)	316 (34.8%)	312 (40.2%)	303 (46.5%)	279 (54.6%)
	68–77 years	634 (50.6%)	606 (54.8%)	571 (56.3%)	510 (56.2%)	424 (54.6%)	332 (50.9%)	223 (43.6%)
	≥78 years	301 (24.0%)	182 (16.5%)	125 (12.3%)	81 (8.9%)	41 (5.3%)	17 (2.6%)	9 (1.8%)
BMI group	Normal (18.5–24.9)	451 (36.0%)	405 (36.6%)	379 (37.4%)	346 (38.2%)	304 (39.1%)	262 (40.2%)	204 (39.9%)
	Overweight (25–29.9)	759 (60.6%)	666 (60.2%)	604 (59.6%)	534 (58.9%)	451 (58.0%)	372 (57.1%)	295 (57.7%)
	Obese (≥30)	43 (3.4%)	35 (3.2%)	31 (3.1%)	27 (3.0%)	22 (2.8%)	18 (2.8%)	12 (2.4%)
ADL	0–4	233 (18.6%)	106 (9.6%)	46 (4.5%)	15 (1.7%)	2 (0.3%)	0 (0.0%)	0 (0.0%)
	5–9	858 (68.5%)	838 (75.8%)	806 (79.5%)	732 (80.5%)	613 (78.9%)	490 (75.2%)	349 (68.3%)
	10–14	104 (8.3%)	104 (9.4%)	104 (10.3%)	104 (11.5%)	104 (13.4%)	104 (16.0%)	104 (20.4%)
	15–20	58 (4.6%)	58 (5.2%)	58 (5.7%)	58 (6.4%)	58 (7.5%)	58 (8.9%)	58 (11.4%)
ABPET	< 2.5	845 (67.4%)	820 (74.1%)	790 (77.9%)	734 (80.9%)	668 (86.0%)	590 (90.5%)	483 (94.5%)
	≥ 2.5	408 (32.6%)	286 (25.9%)	224 (22.1%)	173 (19.1%)	109 (14.0%)	62 (9.5%)	28 (5.5%)
CDRSB	≤10	889 (71.0%)	797 (72.1%)	730 (72.0%)	652 (71.9%)	563 (72.5%)	461 (70.7%)	351 (68.7%)
	> 10	364 (29.1%)	309 (27.9%)	284 (28.0%)	255 (28.1%)	214 (27.5%)	191 (29.3%)	160 (31.3%)
Overall		**1253 (100%)**	**1106 (100%)**	**1014 (100%)**	**907 (100%)**	**777 (100%)**	**652 (100%)**	**511 (100%)**

Table 3: Retention over time by subgroups of sex, age, BMI, ADL, and baseline ABPET and CDRSB. Percentages represent the proportion of the total sample retained at each time point.

Variable	Group	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Sex	Male	616 (49.2%)	583 (52.7%)	551 (54.3%)	506 (55.8%)	463 (59.6%)	409 (62.7%)	341 (66.7%)
	Female	637 (50.8%)	523 (47.3%)	463 (45.7%)	401 (44.2%)	314 (40.4%)	243 (37.3%)	170 (33.3%)
Age group	≤67 years	318 (25.4%)	318 (28.8%)	318 (31.4%)	316 (34.8%)	312 (40.2%)	303 (46.5%)	279 (54.6%)
	68–77 years	634 (50.6%)	606 (54.8%)	571 (56.3%)	510 (56.2%)	424 (54.6%)	332 (50.9%)	223 (43.6%)
	≥78 years	301 (24.0%)	182 (16.5%)	125 (12.3%)	81 (8.9%)	41 (5.3%)	17 (2.6%)	9 (1.8%)
BMI group	Normal (18.5–24.9)	451 (36.0%)	405 (36.6%)	379 (37.4%)	346 (38.2%)	304 (39.1%)	262 (40.2%)	204 (39.9%)
	Overweight (25–29.9)	759 (60.6%)	666 (60.2%)	604 (59.6%)	534 (58.9%)	451 (58.0%)	372 (57.1%)	295 (57.7%)
	Obese (≥30)	43 (3.4%)	35 (3.2%)	31 (3.1%)	27 (3.0%)	22 (2.8%)	18 (2.8%)	12 (2.4%)
ADL	0–4	233 (18.6%)	106 (9.6%)	46 (4.5%)	15 (1.7%)	2 (0.3%)	0 (0.0%)	0 (0.0%)
	5–9	858 (68.5%)	838 (75.8%)	806 (79.5%)	732 (80.5%)	613 (78.9%)	490 (75.2%)	349 (68.3%)
	10–14	104 (8.3%)	104 (9.4%)	104 (10.3%)	104 (11.5%)	104 (13.4%)	104 (16.0%)	104 (20.4%)
	15–20	58 (4.6%)	58 (5.2%)	58 (5.7%)	58 (6.4%)	58 (7.5%)	58 (8.9%)	58 (11.4%)
ABPET	< 2.5	845 (67.4%)	820 (74.1%)	790 (77.9%)	734 (80.9%)	668 (86.0%)	590 (90.5%)	483 (94.5%)
	≥ 2.5	408 (32.6%)	286 (25.9%)	224 (22.1%)	173 (19.1%)	109 (14.0%)	62 (9.5%)	28 (5.5%)
CDRSB	≤10	889 (71.0%)	797 (72.1%)	730 (72.0%)	652 (71.9%)	563 (72.5%)	461 (70.7%)	351 (68.7%)
	> 10	364 (29.1%)	309 (27.9%)	284 (28.0%)	255 (28.1%)	214 (27.5%)	191 (29.3%)	160 (31.3%)
Overall		**1253 (100%)**	**1106 (100%)**	**1014 (100%)**	**907 (100%)**	**777 (100%)**	**652 (100%)**	**511 (100%)**

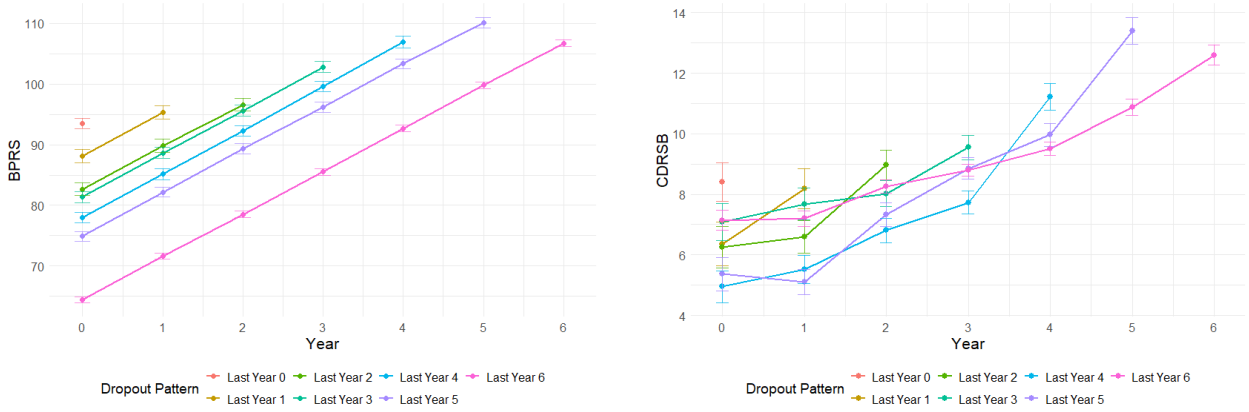


Figure 1: Mean BPRS (left) and CDRSB profiles (right) by dropout pattern. Error bars represent ± 1 standard error.

3.2 Random-effects model for BPRS score

Different covariance structures were evaluated to account for the within-subject correlation over time. Models with unstructured (UN), Toeplitz, and ARH(1) covariance structures were compared using restricted

maximum likelihood estimation. Likelihood ratio tests based on REML indicated that the UN model does not fit significantly better than the Toeplitz model (df = 13, p = 0.65) or ARH1(1) model (df = 8, p = 0.99) (Table 4). Because the ARH(1) structure is the simplest and achieved a comparable fit with fewer parameters, it was selected as the final covariance structure for the random-effects model.

Table 4: Overview of likelihood ratio test results (based on REML) for covariance structure selection.

Model	Covariance	Parameter	-2ℓ	df	G^2	p-value
Model 1	UN	28	28956			
Model 2	Toep	13	28968.4	15	12.4	0.65
Model 3	ARH(1)	8	28968	5	0.4	0.99

Note: Parameters reflect only covariance (random-effects) and D-matrix rank; fixed-effect parameters are not shown.

The most parsimonious LMM was selected through likelihood ratio testing. The full model included all covariates (sex, age, BMI, baseline CDRSB and ABPET) and their interaction with year. Likelihood ratio tests were used to compare nested models, sequentially removing non-significant interactions of covariates with year to simplify the mean structure (Table 5). Under the MAR assumption, the direct likelihood analysis indicated that baseline patient characteristics, including sex, age, BMI, baseline CDRSB and ABPET, were significantly associated with BPRS scores at baseline. In addition, the change in BPRS over time was significantly associated with baseline CDRSB ($\beta = -0.022$; SE = 0.004; $p < 0.0001$). These results rely on the assumption that missingness depends only on observed information; thus, sensitivity analyses are needed to provide greater confidence in the conclusions.

Table 5: Overview of likelihood ratio test results (based on ML) for reduction of preliminary mean structure.

Model	Description	Parameter	-2ℓ	df	G^2	P-value
Full model	All covariates + Time effects	26	28908.5			
Model 1	Full model – baseline ABPET time effect	25	28908.8	1	0.1	0.50
-	Model 1 – baseline CDRSB time effect	24	28936.7	1	19.7	< 0.0001
Model 2	Model 1 – BMI time effect	24	28909.1	1	19.7	0.159
Model 3	Model 2 – age time effect	23	28910.8	1	15.5	0.19
Model 4	Model 3 – sex time effect	21	28911	2	0.1	0.90

Table 6: Parameter estimates for the final random-effects model for BPRS (Direct likelihood)

Effect	**Estimate**	**Standard Error**	**P-value**
Intercept	-70.290	1.406	$< .0001$
Sex (Female)	-0.796	0.182	$< .0001$
Age	1.883	0.020	$< .0001$
BMI	0.143	0.041	0.0006
Baseline CDRSB	-0.031	0.013	0.018
Baseline ABPET	2.694	0.317	$< .0001$
Year	7.224	0.040	$< .0001$
Baseline CDRSB × Year	-0.022	0.004	$< .0001$

3.2.1 Sensitivity analysis

Table 7 summarizes the results of a pattern–mixture–based sensitivity analysis for the continuous BPRS outcome under the ACMV, CCMV, and NCMV identifying restrictions. While small differences in point estimates and standard errors were observed across the three analyses, the estimated effect of time was consistently positive and of similar magnitude under all identifying restrictions. In addition, the interaction

between baseline CDR-SB and time remained negative across all models, indicating slightly slower progression among more impaired patients.

The effects of age, sex, body mass index, and baseline ABPET are stable in direction and statistical significance across the three identifying restrictions. Although minor variations in the magnitude of these effects are present, none of these differences lead to changes in the substantive conclusions. Overall, the similarity of results across ACMV, CCMV, and NCMV suggests that the main longitudinal findings are robust to alternative assumptions about the post-dropout behavior of the outcome.

Table 7: Sensitivity analysis for BPRS based on pattern mixture models with different identifying restrictions (ACMV, CCMV, NCMV).

Effect	**Parameter**	**ACMV**	**CCMV**	**NCMV**
		Estimate (standard error)		
Intercept	β_0	$-70.98\,(1.48)$	$-70.92\,(1.50)$	$-70.15\,(1.26)$
Sex (Female)	β_1	$-0.75\,(0.18)$	$-0.74\,(0.18)$	$-0.79\,(0.11)$
Age	β_2	$1.89\,(0.02)$	$1.89\,(0.02)$	$1.88\,(0.02)$
BMI	β_3	$0.16\,(0.04)$	$0.16\,(0.04)$	$0.14\,(0.03)$
Baseline CDRSB	β_4	$-0.03\,(0.01)$	$-0.03\,(0.01)$	$-0.03\,(0.01)$
Baseline ABPET	β_5	$2.65\,(0.32)$	$2.64\,(0.32)$	$2.71\,(0.19)$
Time (year)	β_6	$7.22\,(0.06)$	$7.23\,(0.06)$	$7.15\,(0.10)$
Baseline CDRSB $\times$ Time	β_7	$-0.02\,(0.01)$	$-0.02\,(0.01)$	$-0.02\,(0.01)$
		P-values		
Intercept	β_0	$< .0001$	$< .0001$	$< .0001$
Sex (Female)	β_1	$< .0001$	$< .0001$	$< .0001$
Age	β_2	$< .0001$	$< .0001$	$< .0001$
BMI	β_3	$< .001$	$< .001$	$< .0001$
Baseline CDR-SB	β_4	0.024	0.022	0.0055
Baseline ABPET	β_5	$< .0001$	$< .0001$	$< .0001$
Time (year)	β_6	$< .0001$	$< .0001$	$< .0001$
Baseline CDRSB $\times$ Time	β_7	$< .0001$	$< .001$	0.0055

3.3 Marginal model for CDRSB

In the GEE analysis from the previous report, sex and BMI were significant predictors of high CDRSB scores, with females having lower odds than males at baseline ($\beta_1 = -0.332$; $e^{-0.332} = 0.717$; p = 0.020) and higher BMI associated with increased odds of high CDRSB ($\beta_3 = 0.065$; $e^{0.065} = 1.067$; p = 0.040). However, when dropout model was accounted for using WGEE, the effects of sex and BMI were no longer significant (sex: $\beta_1 = -0.203$; p = 0.153; BMI: $\beta_3 = 0.054$; p = 0.100) (Table 8). This suggests that the associations observed in the standard GEE were influenced by missing data, as the missingness model showed dropout was related to ADL and previous outcomes, providing evidence against a MCAR missingness mechanism (Table 9). In WGEE the standard errors of the parameter estimates were larger than in GEE for ABPET, year and ABPET $\times$ year, which is common in WGEE because additional sources of uncertainty, due to missingness, are taken into account [3].

The MI-GEE analyses using additional numbers of imputations (15 and 20) showed parameter estimates, standard errors, and P-values similar to those with 10 imputations for all covariates, except for the ABPET and ABPET $\times$ year parameter estimates, reflecting the uncertainty for ABPET also seen in WGEE (Table A1). The P-values for ABPET and ABPET $\times$ year were unaffected and all around 0.8 (Table A1). These results indicate that 10 imputations were sufficient. The standard errors of the parameter estimates were smaller in MI-GEE than WGEE for sex, year, and several of the interactions with year (Table 8). For sex and BMI the parameter estimates were also slightly higher in absolute value compared to WGEE, resulting in a significant association of BMI and the odds of high CDRSB ($\beta_3 = 0.061$; $e^{0.061} = 1.063$; p = 0.041,

Table 8). These differences between results of different models highlight the importance of carefully choosing the most appropriate model for interpretation. As mentioned in the methods section, because of the nature of the study, WGEE was preferred here for interpretation.

The answer to the research question at the population-averaged level was the same for WGEE and MI-GEE: the effects of ADL and ABPET on the odds of high CDRSB were small and non-significant, both at baseline and over time (Table 8).

Table 8: Parameter estimates from GEE and WGEE models, and MI-GEE after 10 multiple imputations (unstructured working correlation).

Effect	**Parameter**	**GEE**			**WGEE**			**MI-GEE**		
		Estimate	Robust SE	P-value	Estimate	SE	P-value	Estimate	SE	P-value
Intercept	β_0	-2.261	1.413	0.110	-2.085	1.428	0.144	-2.837	1.375	0.039
Sex (Female)	β_1	-0.332	0.141	0.020	-0.203	0.142	0.153	-0.246	0.137	0.073
Age	β_2	-0.006	0.018	0.748	0.001	0.018	0.940	0.002	0.017	0.912
BMI	β_3	0.065	0.032	0.040	0.054	0.033	0.100	0.061	0.030	0.041
ADL	β_4	0.002	0.028	0.943	-0.018	0.028	0.530	0.008	0.028	0.772
ABPET	β_5	0.030	0.246	0.904	-0.049	0.283	0.862	0.063	0.235	0.790
Year	β_6	0.808	0.450	0.073	1.020	0.509	0.045	0.909	0.431	0.035
Sex (Female) × Year	β_7	0.081	0.045	0.071	0.036	0.056	0.514	0.063	0.044	0.152
Age × Year	β_8	-0.001	0.006	0.845	-0.006	0.006	0.311	-0.003	0.005	0.591
BMI × Year	β_9	-0.019	0.010	0.070	-0.023	0.013	0.074	-0.018	0.009	0.057
ADL × Year	β_{10}	-0.001	0.009	0.897	0.009	0.010	0.360	-0.006	0.009	0.531
ABPET × Year	β_{11}	-0.009	0.080	0.907	0.056	0.137	0.684	0.016	0.076	0.831

Table 9: Parameter estimates for the missingness model used in WGEE.

Parameter	**Estimate**	**SE**	**P-value**
Intercept	2.118	1.617	0.190
Sex (Female)	-0.078	0.097	0.420
Age	-0.064	0.020	0.001
BMI	-0.002	0.023	0.925
ADL	0.822	0.050	< 0.001
ABPET	-0.028	0.187	0.880
Previous outcome	-1.265	0.099	< 0.001

3.4 Random-effects model for CDRSB

Using the direct likelihood GLMM approach, the results from the second report remain valid under the MAR assumption. Only sex was significantly associated with the longitudinal CDRSB outcome at the subject-specific level (Table 10). Conditional on random intercept and slope, and adjusting for other covariates, with BMI, ADL, and baseline ABPET held constant, female patients had 58% lower baseline odds of having a high CDRSB score compared to male patients. The sex odds ratio increased by nearly 27% per year, meaning the difference in odds between female and male patients became smaller over time. Conditional on random effects and adjusting for other covariates, the baseline odds of a high CDRSB score were not significantly associated with ADL or ABPET. Baseline ADL and ABPET had no significant effect on the annual change in the odds of higher CDRSB, as indicated by the non-significant time interactions for ADL and ABPET (Table 10).

Table 10: Parameter estimates, standard error and P-values for the random-effects model (adaptive Gaussian quadrature with Q=20).

Effect	**Estimate**	**Standard Error**	**P-value**
Intercept	0.184	3.553	0.959
Sex (Female)	-0.849	0.339	0.012
Age	-0.074	0.044	0.090
BMI	0.050	0.077	0.519
ADL	-0.062	0.071	0.383
ABPET	0.797	0.582	0.171
Year	0.212	1.095	0.847
Sex (Female) × Year	0.238	0.106	0.025
Age × Year	0.015	0.013	0.266
BMI × Year	-0.016	0.024	0.518
ADL × Year	0.017	0.022	0.443
ABPET × Year	-0.189	0.184	0.306

4 Concluding remarks

The missingness pattern was monotone, and the exploratory analysis found apparent associations between dropout and several baseline covariates like age and ADL, which is inconsistent with an MCAR missingness mechanism. Mean BPRS and CDRSB trajectories stratified by dropout pattern showed that, at each time point, patients who dropped out at the next checkup exhibited the most severe dementia. Given the longitudinal design, the monotone dropout structure and the dependence of dropout on observed measurements, missingness was assumed to be MAR.

For the continuous BPRS outcome, the likelihood-based linear mixed-effects model identified strong baseline associations and a significant interaction between baseline CDRSB and time. Sensitivity analysis, using the pattern-mixture framework, showed that these conclusions were robust across alternative identifying restrictions (ACMV, CCMV, and NCMV), suggesting that moderate departures from MAR would not alter the substantive interpretation of disease progression on the BPRS scale.

For the binary CDRSB outcome, results were consistent for population-averaged and subject-specific frameworks. Neither ADL not ABPET affected the odds of high CDRSB, either at baseline or over time, regardless of whether marginal (WGEE, MI-GEE) or subject-specific (GLMM) models were used. Associations with sex that appeared significant in standard GEE became non-significant once missing data was handled appropriately using WGEE or multiple imputation. This suggests that earlier findings were most likely driven by informative dropout rather than true population-level effects.

Acknowledgement

We used Gemini and ChatGPT for help to debug SAS code, to generate LaTeX code for formulas and references, and to improve phrasing and grammar.

References

[1] Molenberghs G, Thijs H, Michiels B, Verbeke G, Kenward MG. Pattern-mixture models. Journal de la Société Française de Statistique. 2004;145:49-77.

[2] Osman A, Radman D, Belchior P, Gélinas I. A systematic review of psychometric properties of questionnaires assessing activities of daily living among older adults with neurocognitive disorders. Australian Occupational Therapy Journal. 2025;72:e13013.

[3] Molenberghs G, Verbeke G. Models for Discrete Longitudinal Data. New York: Springer; 2005.

[4] Verbeke G, Molenberghs G. Linear Mixed Models for Longitudinal Data. 2nd ed. New York: Springer; 2009.

[5] Little RJA, Rubin DB. Statistical Analysis with Missing Data. 1st ed. New York: John Wiley Sons; 1987.

[6] Liang CS, Li DJ, Yang FC, Tseng PT, Carvalho AF, Stubbs B, et al. Mortality rates in Alzheimer's disease and non-Alzheimer's dementias: a systematic review and meta-analysis. The Lancet Healthy Longevity. 2021;2(8):e479-88. Available from: https://doi.org/10.1016/S2666-7568(21)00140-9.

Appendices

A Additional Tables

Table A1: Comparison of MI-GEE parameter estimates across 10, 15, and 20 imputations.

Parameter	**10 Imputations**			**15 Imputations**			**20 Imputations**		
	Est.	SE	P-value	Est.	SE	P-value	Est.	SE	P-value
Intercept	-2.837	1.375	0.039	-2.810	1.380	0.042	-2.791	1.376	0.043
Sex (Female)	-0.246	0.137	0.073	-0.244	0.136	0.073	-0.245	0.136	0.071
Age	0.002	0.017	0.912	0.002	0.017	0.915	0.002	0.017	0.922
BMI	0.061	0.030	0.041	0.061	0.030	0.040	0.061	0.030	0.042
ADL	0.008	0.028	0.772	0.007	0.028	0.787	0.008	0.028	0.786
ABPET	0.063	0.235	0.790	0.048	0.237	0.839	0.052	0.236	0.827
Year	0.909	0.431	0.035	0.909	0.431	0.035	0.899	0.432	0.038
Sex (Female) × Year	0.063	0.044	0.152	0.063	0.043	0.145	0.062	0.043	0.142
Age × Year	-0.003	0.005	0.591	-0.003	0.005	0.574	-0.003	0.005	0.590
BMI × Year	-0.018	0.009	0.057	-0.018	0.009	0.053	-0.018	0.009	0.057
ADL × Year	-0.006	0.009	0.531	-0.005	0.009	0.556	-0.005	0.009	0.557
ABPET × Year	0.016	0.076	0.831	0.024	0.076	0.750	0.021	0.075	0.778

B SAS Code

```
/* Code for Random Effects Model for BPRS score*/
proc import out=alzheimerlongas4
          datafile='/home/u64051434/LDA Assignment 3/alzheimerlongas4.txt'
          dbms=dlm replace;
    delimiter='09'x; /* Tab character */
run;

/* full model + UN */
proc mixed data=alzheimerlongas4 method=reml empirical;
    class patid yearclass sex;
    model bprs = sex age bmi cdrsbbl abpetbl
                year sex*year age*year bmi*year cdrsbbl*year abpetbl*year
                / noint solution;
    random intercept year/ type=un subject=patid;
    repeated yearclass / type=un subject=patid;
run;

/* full model + toep */
proc mixed data=alzheimerlongas4 method=reml empirical;
    class patid yearclass sex;
    model bprs = sex age bmi cdrsbbl abpetbl
                year sex*year age*year bmi*year cdrsbbl*year abpetbl*year
```

```
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=toep subject=patid;
run;

/* full model + arh(1) */
proc mixed data=alzheimerlongas4 method=reml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year age*year bmi*year cdrsbbl*year abpetbl*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;


/* now reducing mean structure; from here ML is used*/

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year age*year bmi*year cdrsbbl*year abpetbl*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year age*year bmi*year cdrsbbl*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year age*year bmi*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year age*year cdrsbbl*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year sex*year cdrsbbl*year
               / noint solution;
```

```
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;

proc mixed data=alzheimerlongas4 method=ml empirical;
   class patid yearclass sex;
   model bprs = sex age bmi cdrsbbl abpetbl
               year cdrsbbl*year
               / noint solution;
   random intercept year/ type=un subject=patid;
   repeated yearclass / type=arh(1) subject=patid;
run;


---------------/* sensitivity analysis */----------------

%let SEED = 486048;
%let M = 10; /* number of imputations */
%let LIB = LDA; /* your library */

/*----------------------------------------
  0) Prepare WIDE dataset (BPRS0BPRS6)
----------------------------------------*/
data bprs_wide;
  set &LIB..alzheimer25;
  keep patid sex wzc age adl bmi abpet0 cdrsb0 BPRS0-BPRS6;
run;

proc sort data=bprs_wide; by patid; run;

/*----------------------------------------
  1) Enforce MONOTONE missingness structure
     (Creates bprs_wide_mono with _Imputation_=1..&M)
----------------------------------------*/
proc mi data=bprs_wide
       seed=&SEED
       out=bprs_wide_mono
       nimpute=&M
       round=0.1;
  class sex wzc;
  var sex wzc age adl bmi abpet0 cdrsb0 BPRS0-BPRS6;

  /* Create a monotone missingness form using MCMC */
  mcmc;
run;

/*========================================================
  Utility macro: WIDE -> LONG
========================================================*/
%macro wide_to_long(in=, out=);
data &out;
  set &in;
  array y[7] BPRS0-BPRS6;
  do time = 0 to 6;
    BPRS = y[time+1];
    output;
  end;
  keep _imputation_ patid sex wzc age adl bmi abpet0 cdrsb0 time BPRS;
run;
```

```
proc sort data=&out;
  by _imputation_ patid time;
run;
%mend;

/*=========================================================
  Utility macro: Fit model + MIANALYZE
  (same model everywhere)
=========================================================*/
%macro fit_and_pool(data_long=, out_fixef=, title=);
title "&title";

ods output SolutionF=&out_fixef;

proc mixed data=&data_long method=reml covtest;
  by _imputation_;
  class patid sex (ref='0') wzc;
  model BPRS =
        sex age bmi cdrsb0 abpet0 time cdrsb0*time
        / solution;
  random intercept time / subject=patid type=un;
  repeated time / subject=patid type=arh(1);
run;

title;

proc mianalyze parms=&out_fixef;
  class sex;
  modeleffects Intercept sex age bmi cdrsb0 abpet0 time cdrsb0*time;
run;
%mend;

/*=========================================================
  2) ACMV (benchmark; MAR-compatible in PMM)
     - No MNAR statement needed
=========================================================*/
proc mi data=bprs_wide_mono seed=&SEED out=bprs_wide_acmv nimpute=1;
  by _imputation_;
  class sex wzc;
  var sex wzc age adl bmi abpet0 cdrsb0 BPRS0-BPRS6;

  monotone reg;
run;

%wide_to_long(in=bprs_wide_acmv, out=bprs_acmv_long);

%fit_and_pool(
  data_long=bprs_acmv_long,
  out_fixef=acmv_fixef,
  title=ACMV: Pattern-mixture MI (benchmark)
);

/*=========================================================
  3) CCMV (alternative identifying restriction)
=========================================================*/
proc mi data=bprs_wide_mono seed=&SEED out=bprs_wide_ccmv nimpute=1;
  by _imputation_;
  class sex wzc;
```

```
  var sex wzc age adl bmi abpet0 cdrsb0 BPRS0-BPRS6;

  monotone reg;

  /* CCMV identifying restriction */
  mnar model (BPRS1 BPRS2 BPRS3 BPRS4 BPRS5 BPRS6 / modelobs=ccmv);
run;

%wide_to_long(in=bprs_wide_ccmv, out=bprs_ccmv_long);

%fit_and_pool(
  data_long=bprs_ccmv_long,
  out_fixef=ccmv_fixef,
  title=CCMV: Pattern-mixture MI (departure from ACMV)
);

/*=========================================================
  4) NCMV (alternative identifying restriction)
=========================================================*/
proc mi data=bprs_wide_mono seed=&SEED out=bprs_wide_ncmv nimpute=1;
  by _imputation_;
  class sex wzc;
  var sex wzc age adl bmi abpet0 cdrsb0 BPRS0-BPRS6;

  monotone reg;

  /* NCMV identifying restriction */
  mnar model (BPRS1 BPRS2 BPRS3 BPRS4 BPRS5 BPRS6 / modelobs=ncmv);
run;

%wide_to_long(in=bprs_wide_ncmv, out=bprs_ncmv_long);

%fit_and_pool(
  data_long=bprs_ncmv_long,
  out_fixef=ncmv_fixef,
  title=NCMV: Pattern-mixture MI (departure from ACMV)
);

/* ----------Code for WGEE---------------- */
data for_gee;
    set alzheimer25_long;
    by patid;

    /* Create the lagged variable */
    previous_outcome = lag(cdrsbb);

    if first.patid then previous_outcome = 1;
run;

proc gee data=for_gee descending;
    class patid yearclass sex (ref = '0');
    model cdrsbb = sex bmi age adl abpetbl year
                  sex*year bmi*year age*year adl*year abpetbl*year / dist=bin link=logit;
    repeated subject=patid / type=unstr;

    /* the model of missingness */
    missmodel sex adl abpetbl bmi age previous_outcome / type=obslevel;
run;
```

```
/* --------------Code for MI-GEE------------------ */

/* Data imputation with M=10, 15 and 20 (M=10 shown here);
Imputed original data, not the binary version for more precision; */

proc mi data=datasets.alzheimerhoras3 seed=486048 simple out=datasets.alzheimerhoras3a nimpute=10
    round=1;
var cdrsb0 cdrsb1 cdrsb2 cdrsb3 cdrsb4 cdrsb5 cdrsb6
       sex bmi age adl abpet0;
run;

/* dichotomization of imputed data*/

data datasets.alzheimerhoras3b;
set datasets.alzheimerhoras3a;
cdrsbb0=1; if cdrsb0 <= 10 then cdrsbb0=0;
cdrsbb1=1; if cdrsb1 <= 10 then cdrsbb1=0;
cdrsbb2=1; if cdrsb2 <= 10 then cdrsbb2=0;
cdrsbb3=1; if cdrsb3 <= 10 then cdrsbb3=0;
cdrsbb4=1; if cdrsb4 <= 10 then cdrsbb4=0;
cdrsbb5=1; if cdrsb5 <= 10 then cdrsbb5=0;
cdrsbb6=1; if cdrsb6 <= 10 then cdrsbb6=0;
if cdrsb0=. then cdrsbb0=.;
if cdrsb1=. then cdrsbb1=.;
if cdrsb2=. then cdrsbb2=.;
if cdrsb3=. then cdrsbb3=.;
if cdrsb4=. then cdrsbb4=.;
if cdrsb5=. then cdrsbb5=.;
if cdrsb6=. then cdrsbb6=.;
run;

/* transforming horizontal dataset in vertical dataset
and adding the yearclass variable we need later
 */
data datasets.alzheimerhoras3c;
set datasets.alzheimerhoras3b;

array x (7) cdrsbb0 cdrsbb1 cdrsbb2 cdrsbb3 cdrsbb4 cdrsbb5 cdrsbb6;
array y (7) cdrsb0 cdrsb1 cdrsb2 cdrsb3 cdrsb4 cdrsb5 cdrsb6;

do j = 1 to 7;
   cdrsbb = x(j); /* binary outcome */
   cdrsb = y(j); /* continuous outcome */
   year = j - 1; /* 06 */
   yearclass = year; /* categorical time */
   output;
end;

drop j cdrsbb0-cdrsbb6 cdrsb0-cdrsb6;
run;

/* sorting to make sure gee works correctly with years in right order */
proc sort data=datasets.alzheimerhoras3c;
    by _Imputation_ patid yearclass;
run;

/* the analysis task (GEE) */
proc gee data=datasets.alzheimerhoras3c DESCENDING;
       title "GEE after multiple imputation";
```

```
       class patid yearclass sex (ref='0');
       by _imputation_;
       model cdrsbb = sex bmi age adl abpet0 year
                 sex*year bmi*year age*year adl*year abpet0*year
                 / dist=binomial;
   repeated subject=patid / withinsubject=yearclass type=unstr covb corrw modelse;
       ods output
              GEEEmpPEst = datasets.gmparms
       GEERCov = datasets.gmcovb
       ParmInfo = datasets.gmpinfo;
run;

/* Extra steps needed to prepare for MIANALYZE because parameter names
not the same and 2 levels of sex divided over different columns */

data gmparms_mi;
  set datasets.gmparms;
  length Parameter $32;
  if Parm = 'sex' then Parameter = cats('sex', Level1);
  else if Parm = 'year*sex' then Parameter = cats('year_sex', Level1);
  else if Parm = 'abpetbl' then Parameter = 'abpet0';
  else if Parm = 'bmi*year' then Parameter = 'bmi_year';
  else if Parm = 'age*year' then Parameter = 'age_year';
  else if Parm = 'adl*year' then Parameter = 'adl_year';
  else if Parm = 'abpet0*year' then Parameter = 'abpet0_year';
  else Parameter = Parm;

  /*Drop the reference level rows (ref=0) */
  if (Parm = 'sex' or Parm = 'year*sex') and Level1 = '0' then delete;

  keep _Imputation_ Parameter Estimate StdErr;
run;

/* the inference task */

proc mianalyze parms=gmparms_mi;
   modeleffects Intercept
               sex1 bmi age adl abpet0 year
               year_sex1 bmi_year age_year adl_year abpet0_year;
run;


/* ---------------------------------------
GLMM for analysis of binary CDRSB
-----------------------------------------*/

proc nlmixed data=datasets.alzheimer25_long qpoints=20;

   parms
      beta0 = 0.26
      beta_sex = -0.71
      beta_bmi = 0.05
      beta_age = -0.08
      beta_adl = -0.05
      beta_abp = 1.02
      beta_yr = 0
      beta_sex_yr = 0.19
      beta_bmi_yr = -0.013
      beta_age_yr = 0.019
```

```
      beta_adl_yr = 0.017
      beta_abp_yr = -0.26
      d11 = 18
      d12 = 4
      d22 = 1.8
   ;

   /* Linear predictor */
   teta = beta0
        + b1
        + beta_sex * sex
        + beta_bmi * bmi
        + beta_age * age
        + beta_adl * adl
        + beta_abp * abpetbl

        + b2 * year
        + beta_yr * year
        + beta_sex_yr * sexyear
        + beta_bmi_yr * bmiyear
        + beta_age_yr * ageyear
        + beta_adl_yr * adlyear
        + beta_abp_yr * abpetbly
   ;

   /* Normal likelihood for outcome */
   expteta = exp(teta);
   p = expteta/(1+expteta);
   model cdrsbb ~ binary(p);

   /* Random intercept and slope*/
   random b1 b2 ~ normal([0, 0] , [d11, d12, d22])
   subject=patid;

run;
```